Intercostal cryoanalgesia for acute pain after video-assisted thoracic surgery lung resection: A randomized controlled preliminary trial

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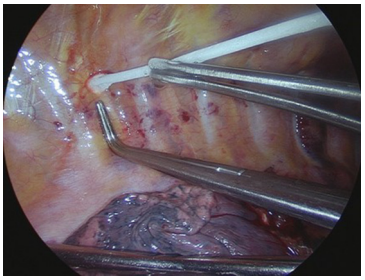
ABSTRACT

Objective: Video-assisted thoracoscopic surgery (VATS) is associated with significant postoperative pain. Multimodal analgesia, including single-shot paravertebral blocks, is widely used but provides limited analgesic duration. Intercostal cryoanalgesia, which offers prolonged pain relief, presents a promising adjunctive option. This study aimed to assess the analgesic benefit of intercostal cryoanalgesia in VATS lung cancer surgery. The primary outcome was thoracic pain during cough 24 hours postsurgery, measured via a verbal numerical rating scale.

Methods: In a randomized, double-blind, controlled trial, 80 patients undergoing VATS lobectomy for lung cancer were assigned to either a control group receiving standard multimodal analgesia with single-shot paravertebral blocks or a cryoanalgesia group receiving additional transpleural intercostal cryoanalgesia (Cryoprobe; Erbe). Thoracic pain at rest and during cough was evaluated at multiple time points up to 6 months postoperatively. Secondary outcomes included quality of recovery, oral morphine equivalents consumption, side effects, thoracic sensory loss (Von Frey filament), and neuropathic pain (Douleur neuropathique 4 score).

Results: Thoracic verbal numerical rating scale during cough at 24 hours showed no significant difference between groups (4.7 ± 2.7 vs 4.8 ± 2.9; $P = .78$). Pain scores, quality of recovery, opioid consumption, side effects, Douleur neuropathique 4 scores, and sensory loss incidence were comparable over 7 days. Pain scores during cough were significantly higher 1 month after surgery (4.7 ± 2.4 vs 3.4 ± 2.0; $P = .036$) but not at 3-month and 6-month follow-up.

Conclusions: In this double-blind trial, measurable cryoanalgesia was not observed in the treatment group during cough at 24 hours post-VATS, nor was measurable thoracic sensory loss or improved acute or chronic recovery outcomes. (J Thorac Cardiovasc Surg 2025;170:1242-51)



Thoracoscopic view showing intercostal cryoanalgesia using the Erbe cryoablation probe.

CENTRAL MESSAGE

Measurable cryoanalgesia was not observed in the treatment group for acute and chronic thoracic pain following VATS lung resection.

PERSPECTIVE

This study suggests that the studied approach did not produce cryoanalgesia nor enhance pain relief after VATS lung resection compared with standard methods. Further research is needed to reliably produce Wallerian degeneration and determine its role in clinical practice.

See Commentary on page 1252.

Video-assisted thoracoscopic surgery (VATS) is associated with a high incidence of moderate to severe acute postoperative pain.[1] Effective pain management during the postoperative period is crucial because it not only facilitates recovery but also enhances patient satisfaction and mitigates the risk of postoperative complications.[2] Recent guidelines from the PROSPECT (procedure specific postoperative pain management) initiative recommend a multimodal analgesia approach for VATS, including regional techniques such as single-shot paravertebral blocks (SPVBs).[3]

PVBs are effective for managing pain in thoracic surgery; however, the analgesic effects of SPVBs are short-lived,

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Abbreviations and Acronyms

DN4 = Douleur neuropathique 4
IV-PCA = intravenous patient-controlled analgesia
PACU = postanesthesia care unit
QoR = quality of recovery
SPVBs = single-shot paravertebral blocks
VATS = video-assisted thoracoscopic surgery
VNRS = verbal numerical rating scale

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typically lasting only 6 to 24 hours.[4,5] With increasing hospital discharge within 24 to 48 hours following surgery using Enhanced Recovery After Surgery protocols, there is a need for prolonged, safe, and effective analgesic modalities. Cryoanalgesia targeting intercostal nerves presents a promising alternative, potentially offering extended pain relief after PVBs have worn off.

Cryoanalgesia entails the application of cold to targeted nerve fibers, achieving temperatures between −20 and −100 °C and inducing reversible interruption of nerve conduction through Wallerian degeneration.[6] Effective analgesia in the affected area persists for several weeks to months.[7] By potentially extending optimal pain control without significant hemodynamic or respiratory side effects, cryoanalgesia could support Enhanced Recovery After Surgery protocols and timely discharge. Although cryoanalgesia has proven benefits in pain management,[6-8] there is limited research specifically assessing its effectiveness for pain control and opioid-sparing in VATS.[9,10] The objective of this randomized controlled preliminary trial was to evaluate the efficacy of a cryoanalgesia technique in the management of acute and chronic thoracic pain after VATS anatomic pulmonary resection for lung cancer.

METHODS

Study Design and Randomization

This randomized, controlled, double-blinded study was approved by the Scientific and Ethics Committees of the Centre hospitalier de l'Université de Montreal (internal review board Nos. 23.138 approved October 23, 2023, and 23.146 approved October 4, 2023, and registered at ClinicalTrials.gov (NCT05896150 registered May 19, 2023, and NCT05884099 registered May 20, 2023). After providing written informed consent, patients undergoing anatomical lung resection by VATS at the Centre hospitalier de l'Universite de Montreal were enrolled from November 2023 to October 2024. Patients were randomized in 2 groups: multimodal analgesia, including SPVB and intercostal cryoanalgesia (cryoanalgesia group) or the same multimodal analgesia protocol, including SPVB without cryoanalgesia (control group).

Patients older than age 18 years with an American Society of Anesthesiologists physical status score of II or III undergoing elective VATS anatomic pulmonary resection (segmentectomy, lobectomy, or bilobectomy) for lung cancer were screened. Exclusion criteria were contraindication to the PVB or to intercostal cryoanalgesia; known allergy to celecoxib, sulfa, or both hydromorphone and morphine; estimated glomerular filtration rate <60 mL/min/1.73 m^2; preoperative thoracic or shoulder pain or history of thoracic surgery on the operated side; high-risk of conversion to thoracotomy as assessed by the surgeon; choice of epidural analgesia; or regular use of opioids or neuropathic pain medication), inability to consent, pregnancy, and refusal to participate.

Patients were contacted by telephone on the day before surgery by a member of the research team using a preapproved script. Written informed consent was obtained on the day of surgery, and preoperative quality of recovery (QoR-15) (Figure 1) questionnaires were completed.

After lung resection and before chest closure, patients were randomized by a research nurse into 2 groups using an independently generated, computer-based allocation sequence (1:1 ratio, variable block sizes). After the attending surgeon had exited the operating room, group assignment was disclosed to the cryoanalgesia operator (E.K.). To maintain blinding, the cryoanalgesia operator oversaw chest closure and patient management until emergence from general anesthesia, and did not participate in postoperative data collection or pain management. The attending surgeon, research investigators, postoperative nursing staff, and patients remained blinded to group allocation throughout the entire follow-up period.

Standardized Perioperative Protocol

All patients received preoperative oral acetaminophen 975 mg and celecoxib 200 mg (100 mg if age 65 years or older or weight <50 kg). Induction agents, including propofol, remifentanil, sufentanil, fentanyl, and rocuronium were administered at the discretion of the attending anesthesiologist. Maintenance was achieved with propofol titrated to maintain bispectral index, Covidien or entropy (Datex-Ohmeda) values between 40 and 60, or 0.7 to 1.2 MAC of desflurane or sevoflurane. Nitrous oxide, ketamine, dexmedetomidine, and meperidine were not permitted. Boluses of fentanyl, sufentanil, or remifentanil were titrated as needed to maintain hemodynamic parameters within 20% of baseline preoperative values. Before emergence, patients received 0.5 to 1 mg subcutaneous hydromorphone (2.5 to 5 mg subcutaneous morphine in cases of hydromorphone intolerance).

Prophylactic antiemetic management consisted of intravenous dexamethasone 0.1 mg/kg (maximum, 8 mg) after induction and intravenous ondansetron 4 mg before emergence. In patients with an Apfel score ≥3, intravenous metoclopramide 10 mg and/or haloperidol 1 mg were allowed. Monitoring included noninvasive and invasive blood pressure, pulse oximetry, 5-lead electrocardiogram, end-tidal carbon dioxide, temperature assessment, and train-of-4 stimulation.

Patient positioning was standardized in the lateral decubitus position, with the operating table maximally flexed at the xiphoid level. The surgical-side shoulder was maintained at 90° of flexion without abduction, with the elbow at 45° of flexion. During surgery, the anatomical resections were carried out using a standardized 3-port VATS technique, except for 1 surgeon (B.N.), who employed a 4-port approach.

In both groups, the paravertebral block was performed by the surgeon after thoracoscope placement and before pulmonary and lymph node resection. The paravertebral space was identified using the thoracoscope. The T5 spinous process was palpated, and the transverse process was estimated approximately 2.5 cm laterally on the operative side. A 17G Tuohy needle (Teleflex) connected to a saline-filled syringe was inserted angled slightly cranially, until a loss of resistance was felt and paravertebral space dilation was confirmed via the thoracoscope with 1 to 2 mL saline without breaching the parietal pleura. If saline diffusion was in an intercostal space the needle was redirected. Once proper needle positioning and craniocaudal saline spread were confirmed, a fractionated injection of bupivacaine 0.5% with epinephrine 5 μg/mL was administered at a dose of 0.4 mL/kg (maximum 40 mL) of the adjusted body weight[11] or actual body weight, whichever was lower,

THOR

QoR-15 Patient Survey

How have you been feeling in the last 24 hours?

(0 to 10, where: 0 = none of the time [poor] and 10 = all of the time [excellent])

1. Able to breathe easily	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
2. Been able to enjoy food	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
3. Feeling rested	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
4. Have had a good sleep	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
5. Able to look after personal toilet and hygiene unaided	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
6. Able to communicate with family or friends	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
7. Getting support from hospital doctors and nurses	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
8. Able to return to work or usual home activities	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
9. Feeling comfortable and in control	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time
10. Having a feeling of general well-being	None of the time	0	1	2	3	4	5	6	7	8	9	10	All of the time

A

Have you had any of the following in the last 24 hours?

(10 to 0, where: 10 = none of the time [excellent] and 0 = all of the time [poor])

11. Moderate pain	None of the time	10	9	8	7	6	5	4	3	2	1	0	All of the time
12. Severe pain	None of the time	10	9	8	7	6	5	4	3	2	1	0	All of the time
13. Nausea or vomiting	None of the time	10	9	8	7	6	5	4	3	2	1	0	All of the time
14. Feeling worried or anxious	None of the time	10	9	8	7	6	5	4	3	2	1	0	All of the time
15. Feeling sad or depressed	None of the time	10	9	8	7	6	5	4	3	2	1	0	All of the time

B

FIGURE 1. Quality of Recovery (QoR-15) patient survery.

observing continuous paravertebral space dilation and ensuring no return of blood during intermittent aspiration.

In the postanesthesia care unit (PACU), intravenous hydromorphone 0.2 to 0.4 mg (or morphine 1 to 2 mg in case of allergy to hydromorphone) was administered every 5 minutes until a verbal numerical rating scale (VNRS) (0-10) pain score <4 was achieved. Once the target VNRS score was reached, patients were transitioned to an intravenous patient-controlled analgesia (IV-PCA) pump set to deliver 0.2 mg hydromorphone (or 1 mg

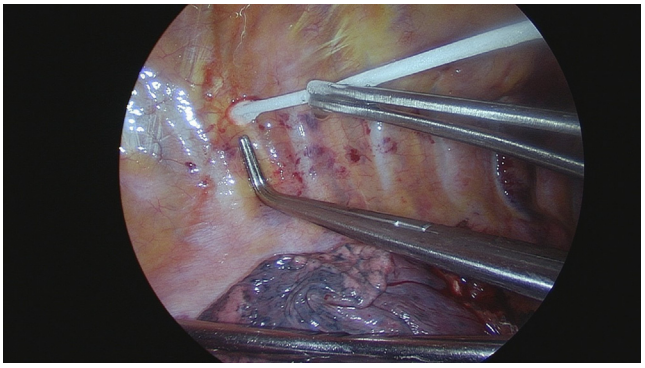

FIGURE 2. Thoracoscopic view showing the ice ball formed at the tip of the Erbe cryoablation probe (*white*) held in position by a video-thoracoscopic thoracic surgery (VATS) Duval forceps. A VATS right-angle dissector with a 3-cm tip is used to estimate the distance from the spine. The lung is retracted from the cryoanalgesia field for security.

morphine) at 5-minute intervals, with a maximum of 10 doses per hour and no continuous basal infusion.

After surgery, patients received scheduled oral acetaminophen 975 mg every 6 hours for 5 days, and oral celecoxib 200 mg (100 mg if older than age 65 years or weight <50 kg) every 12 hours for the first 72 hours and then as needed for up to a total of 7 days. IV-PCA was maintained for up to 24 hours postoperatively, after which hydromorphone, either subcutaneously at doses of 0.5 to 1 mg or orally at 1 to 2 mg, or morphine, either subcutaneously at 2.5 to 5 mg or orally at 5 to 10 mg, was administered every 3 hours as needed.

Standardized Intercostal Cryoanalgesia

For patients allocated to the cryoanalgesia group, after pulmonary resection and before closure of the thoracoscopic ports, a single-use cryoanalgesia probe (Cryoprobe 2.4 mm, 1.15 m, Erbe) was introduced through the utility incision. The probe was positioned 3 cm lateral to the spine using a right-angle instrument with a 3-cm tip for precise and reproducible measurement (Figure 2). It was targeted at the lower border of the third rib, placed directly perpendicular to the nerve when visible through the pleura, while tenting the pleura beneath the rib ridge. The pleura was not dissected or breached, and the nerve was not manipulated. Once the probe tip was correctly positioned and surrounding structures were safely displaced, cryoanalgesia was initiated at −50 to −70 °C for 2 minutes. Following treatment, a minimal volume of warmed saline was applied to the targeted site to facilitate thawing of superficial tissues and allow for safe probe removal. This standardized procedure[6] was subsequently repeated for each intercostal space from T3 to T9, covering a total of 7 levels.

Outcome Measures

The primary outcome was the thorax VNRS during cough 24 hours postsurgery. Secondary outcomes included the assessment of acute postoperative pain severity, measured using the VNRS for pain at rest and during cough at both the thorax and the surgical-side shoulder at 1, 6, 12, 18, 24, and 48 hours postoperatively (documented by recovery room or ward registered nurses while respecting patient sleep) and at 7 days after surgery. Nausea or vomiting, pruritus, and Ramsay sedation scores were documented at the same time points. The incidence of persistent postoperative pain (defined as any pain at rest or during cough) and VNRS scores for persistent thoracic pain at rest and during cough were assessed at 1, 3, and 6 months after surgery. The total amount of intraoperative fentanyl equivalents was recorded, using 10:1 conversion ratio for sufentanil. Remifentanil was recorded separately. Oral morphine equivalents received within the first 24 hours, including preemergence, PACU, and IV-PCA doses, along with the total daily doses on day 2 and day 7, were recorded. The conversion ratio used for hydromorphone to morphine was 1:5. For patients who were discharged before these time points, follow-up telephone calls were conducted to document the amount of opioid consumed in the preceding 24 hours. Patients were discharged with a prescription of oral hydromorphone 1 mg to be taken as needed every 4 hours. If 1 mg was insufficient for pain control, the dose could be increased to 2 mg as directed. Opioid and neuropathic pain medication use at 1, 3, and 6 months was documented. Hypoesthesia was evaluated on the surgical-side thorax using a Von Frey monofilament (6.1 g, Bioseb) at the midclavicular and midscapular lines, with the level of sensory change assessed 24 hours and 1 month postsurgery. Sensory change was deemed significant if it involved >1 dermatome. Postoperative recovery quality was assessed using the QoR-15 questionnaire,[12,13] and the presence of neuropathic pain was evaluated using the Douleur Neuropathique 4 (DN4) score (Figure 3).[14] Both assessments were conducted 48 hours postsurgery by a member of the research team. DN4 assessment was repeated at 1 month during the follow-up visit and again at 6 months via telephone in patients reporting persistent pain. At 6 months, if a patient scored at least 1 point on the interview section of the DN4 questionnaire, they were instructed to perform a self-examination over the telephone, given the possibility of a positive result (≥4 points). The total duration of hospitalization, from PACU arrival to hospital discharge, was recorded. At day 7, 3 months, and 6 months postoperatively, data were collected through a standardized telephone call using a predetermined script. At 1 month, data were collected in person during the surgical follow-up visit, or by telephone if an in-person visit was not feasible.

Sample Size Calculation and Statistical Analysis

Based on a prior study (NCT03860480) from our research team,[1] the mean VNRS for thoracic pain during cough 24 hours post-VATS was 5 ± 2.5. Using this standard deviation and an alpha level of 5%, we calculated that a sample size of 68 patients (34 per group) would yield 90% power to detect a clinically significant reduction of 2 or more VNRS points at 24 hours postsurgery. To accommodate potential losses to follow-up, a total of 80 patients were recruited.

Statistical analysis was performed using R software for Windows 11, version 4.4.2 (R Foundation for Statistical Computing). Demographic and surgical characteristics were compared between groups and assessed for imbalances using the standardized mean difference. The normality of distribution was evaluated by visual inspection of histograms. We compared the acute (up to 7 days) and chronic (at 1, 3, and 6 months) thorax VNRS and postoperative (and delta) QoR-15 scores with Welch *t* test and expressed as mean ± standard deviation. The shoulder VNRS, oral morphine equivalent consumption, length of hospital stay, and chest tube duration were compared using the Mann-Whitney *U* test, with results reported as medians and interquartile ranges. The proportion of patients exhibiting Von Frey hypoesthesia at 24 hours and 1 month, positive DN4 score, postoperative nausea or vomiting, pruritus, persistent postoperative pain, and consumption of opioid or neuropathic pain medication at 1, 3, and 6 months, and the Ramsay Sedation Scale scores were compared using Fisher exact test. All analyses were conducted based on the assigned group following an intention-to-treat approach.

RESULTS

Two hundred thirty-eight patients were screened for eligibility (Figure E1), and 80 were randomized. All patients completed the study. In the cryoanalgesia group, 1 patient did not receive the allocated intervention due to equipment failure, 1 required a repeat VATS at 4 hours for hemostasis, and 2 required thoracic epidurals and thoracotomy for

THOR

DN4 – QUESTIONNAIRE

To estimate the probability of neuropathic pain, please answer yes or no for each item of the following four questions.

INTERVIEW OF THE PATIENT		
QUESTION 1:		
Does the pain have one or more of the following characteristics?	**YES**	**NO**
Burning	☐	☐
Painful cold	☐	☐
Electric shocks	☐	☐
QUESTION 2:		
Is the pain associated with one or more of the following symptoms in the same area?	**YES**	**NO**
Tingling	☐	☐
Pins and needles	☐	☐
Numbness	☐	☐
Itching	☐	☐

EXAMINATION OF THE PATIENT		
QUESTION 3:		
Is the pain located in an area where the physical examination may reveal one or more of the following characteristics?	**YES**	**NO**
Hypoesthesia to touch	☐	☐
Hypoesthesia to pinprick	☐	☐
QUESTION 4:		
In the painful area, can the pain be caused or increased by:	**YES**	**NO**
Brushing?	☐	☐

YES = 1 point
NO = 0 points

Patient's Score: /10

FIGURE 3. Douleur neuropathique 4 (DN4) questionnaire.

bronchopleural fistula at 5 and 7 days. No causal relationship between the intervention and surgical complications was identified. All patients were included in the analyses, with no complications related to cryoanalgesia.

A significant imbalance (standardized mean difference $\geq$0.2) in baseline characteristics, including body mass index and smoking status, was identified (Table 1). Multivariable linear regression adjusting for these variables showed no significant associations with thorax VNRS during cough at 24 hours ($P > .05$). The presence of a preoperative diagnosis of depression, anxiety, or preoperative consumption of related antidepressant medication and preoperative neoadjuvant treatment were similar. The longer anesthesia and surgery durations observed in the cryoanalgesia group are likely attributable to the additional time required to perform the cryoanalgesia procedure (Table 2).

Thorax VNRS during cough at 24 hours showed no significant difference between the control and cryoanalgesia groups (4.7 $\pm$ 2.7 vs 4.8 $\pm$ 2.9; $P = .78$). Similarly, no significant differences were observed at 1 hour, 6 hours, 12 hours, 18 hours, 48 hours, and 7 days postsurgery (Figure 4). Pain scores at rest for the thorax (Figure 5) and at rest and during cough for the shoulder were also comparable across all time points. Median shoulder pain at rest and during cough was 0 in both groups for all measurements. The incidence of persistent thoracic pain was not significantly different in the control and cryoanalgesia groups at 1 (62.5% vs 77.5%; $P = .22$), 3 (30% vs 47.5%; $P = .17$), and 6 months (17.5% vs 20%; $P = 1.0$). In patients with persistent thoracic pain, thorax VNRS during cough at 1 month were significantly higher in the cryoanalgesia group (4.7 $\pm$ 2.4 vs 3.4 $\pm$ 2.0; $P = .036$), but not at rest (3.1 $\pm$ 2.5 vs 2.0 $\pm$ 1.8; $P = .055$). At 3 months, thorax VNRS were similar between groups both at rest (2.6 $\pm$ 2.0 vs 1.9 $\pm$ 1.5; $P = .265$) and during cough (3.7 $\pm$ 2.1 vs 3.0 $\pm$ 1.6; $P = .273$). At 6 months, scores also remained comparable at rest (1.6 $\pm$ 1.8 vs 2.7 $\pm$ 3.1; $P = .442$) and during cough (3.2 $\pm$ 2.7 vs 4.9 $\pm$ 3.1; $P = .309$). No significant differences were observed in

TABLE 1. Baseline characteristics

Variables	Control group (n = 40)	Cryoanalgesia group (n = 40)	SMD
Age (y)	67 ± 11	68 ± 8	0.054
Sex			
Female	19 (47.5)	20 (50)	0.050
Male	21 (52.5)	20 (50)	
BMI	29.2 ± 5.4	25.2 ± 3.9	0.853
ASA classification			
2	23 (57.5)	24 (60.0)	0.051
3	17 (42.5)	16 (40.0)	
Comorbidities			
Active smoking	3 (7.5)	7 (17.5)	0.306
Diabetes	7 (17.5)	8 (20.0)	0.064
Preoperative pain	4 (10.0)	5 (12.5)	0.079
Anxiety/depression diagnosis or medication	8 (20)	6 (15)	0.130
Neoadjuvant therapy	3 (7.5)	5 (12.5)	0.165

Values are presented as mean ± standard deviation or as n (%). *SMD*, Standardized mean difference; *BMI*, body mass index; *ASA*, American Society of Anesthesiologists.

preoperative, postoperative, or delta QoR-15 scores, the incidence of postoperative nausea and vomiting and pruritus at any assessed time points, or the incidence of a positive DN4 score (defined as a score of 4 or more out of 10) (Table 2). Similarly, Ramsay Sedation Scale scores were comparable across all measurements.

Median daily oral morphine equivalents consumption was similar between the control and cryoanalgesia groups within the first 24 hours (64 ± 34.75 vs 66 ± 44 mg; $P = .52$), on day 2 (22.5 ± 30 vs 15 ± 21.3 mg; $P = .08$), and on day 7 (5 ± 10 vs 10 ± 20 mg; $P = .07$). Persistent opioid use at day 7 (52.5% vs 62.5%;

TABLE 2. Comparison of perioperative and postoperative characteristics between groups

Variables	Control group	Cryoanalgesia group	*P* value
Perioperative			
Type of surgery			
Segmentectomy	12 (30)	7 (17.5)	.2933*
Lobectomy	28 (70)	33 (82.5)	
Anesthesia duration (min)	140 ± 31.25	152.5 ± 34	.0490†
Surgery duration (min)	89.5 ± 28.5	106 ± 37	.0176†
No. of thoracoscopy ports	3 ± 0	3 ± 0	.5346†
Fentanyl equivalents during surgery (μg)	225 ± 62.5	200 ± 100	.5135†
Remifentanil during surgery (μg)	110 ± 135	75 ± 80	.0423†
Postoperative			
Nausea or vomiting			
24 h	14 (35)	14 (35)	1.000*
48 h	1 (2.5)	5 (12.5)	.200*
Day 7	3 (7.5)	2 (5)	1.000*
Pruritus			
24 h	2 (5)	6 (15)	.263*
48 h	3 (7.9)	4 (10)	1.000*
Day 7	1 (2.5)	2 (5.1)	.615*
Positive DN4 score			
48 h	4 (10)	8 (20)	.372*
1 mo	11 (27.5)	14 (35)	.115*
6 mo	3 (7.5)	6 (15)	.482*
QoR-15 score			
Preoperative	138.5 ± 15.25	137 ± 18.5	.657†
Postoperative	115.62 (16.62)	117.72 (18.25)	.596‡
Delta§	−21.38 (14.56)	−15.82 (16.45)	.117‡

Values are presented as mean ± standard deviation, median (interquartile range), or as n (%). *DN4*, Douleur neuropathique 4; *QoR-15*, quality of recovery. *Fisher exact test. †Mann-Whitney *U* test. ‡*t* Test. §Postoperative – preoperative.

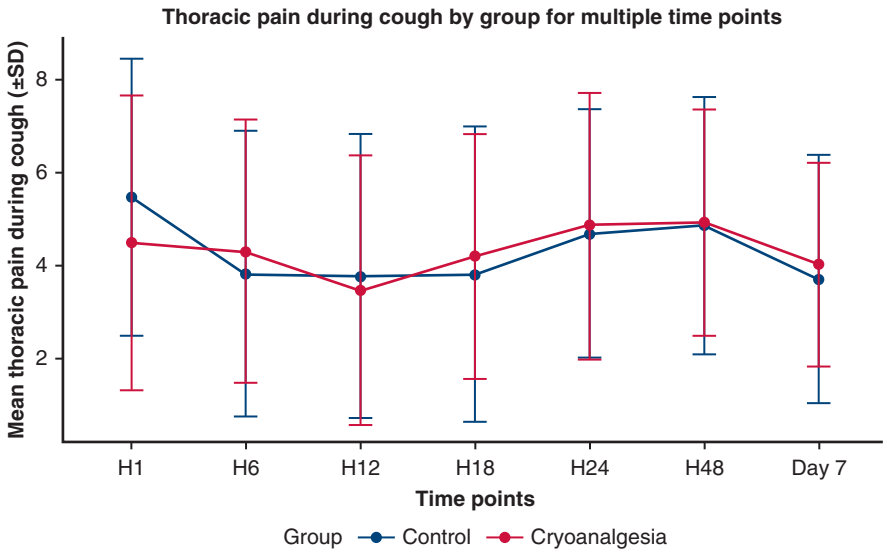

FIGURE 4. Mean acute thoracic verbal numerical rating scale (0-10) during cough across time points by group. *SD*, Standard deviation.

$P = .07$), 1 month (7.5% vs 20%; $P = .115$), 3 months (0% vs 5%; $P = .494$), and 6 months (0% for both groups) were not significantly different. Neuropathic medication use at 1 (2.5% vs 12.5%; $P = .201$), 3 (5% vs 15%; $P = .263$), and 6 months (7.5% vs 12.5%; $P = .712$) were not significantly different. Hospital length of stay (3.0 vs 2.9 days; $P = .83$) and chest tube duration (2.97 vs 4.45 days; $P = .498$) were comparable.

The incidence of positive Von Frey hypoesthesia at 24 hours was similar: 56.4% (22 out of 39) in the control group and 50% (20 out of 40) in the cryoanalgesia group ($P = .65$). Notably, 30% of patients in each group ($P = 1.0$) exhibited hypoesthesia spanning at least 3 dermatomal levels, confirmed at both the midclavicular and midscapular lines. Three patients (1 in the control group, 2 in the cryoanalgesia group) demonstrated hypoesthesia across the expected 7 intercostal dermatomes on Von Frey examination. The incidence of positive Von Frey hypoesthesia at 1 month was also similar between the groups (17 out of 31 [54.8%] vs 16 out of 31 [51.6%]; $P = 1.0$).

DISCUSSION

In this study, transpleural application of the Erbe Cryoprobe (−50 to −70 °C) to 7 intercostal levels did not result in measurable cryoanalgesia. Cryoanalagesia group members did not display better thoracic pain relief during cough at 24 hours post-VATS compared with multimodal analgesia with a SPVB, and also did not exhibit reduced thoracic pain

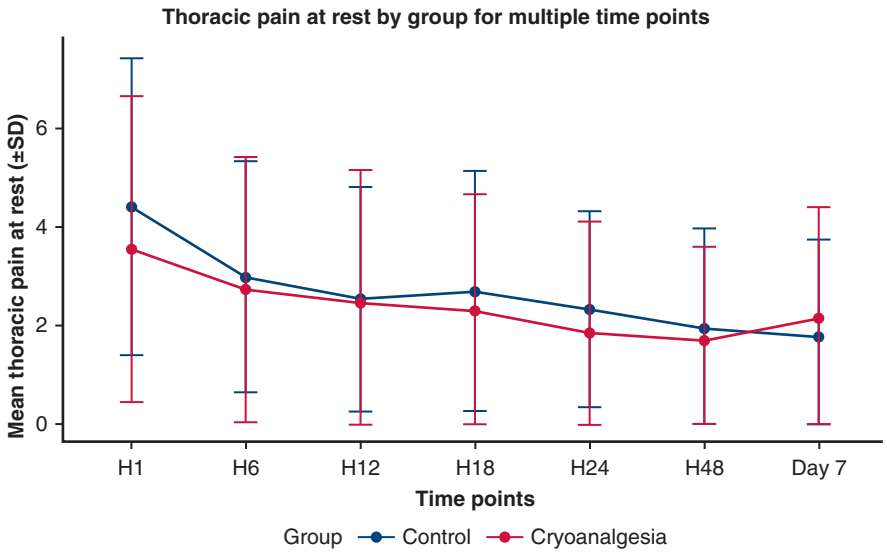

FIGURE 5. Mean acute thoracic verbal numerical rating scale (0-10) at rest across time points by group. *SD*, Standard deviation.

scores at rest or during cough on days 1, 2, and 7, reduced opioid consumption and associated side effects, hospital stay, or incidence of neuropathic pain, or improved quality of recovery. Cryoanalgesia group members even displayed slightly but significantly more pain at cough 1 month postoperatively, a difference that did not persist at the 3- and 6-month evaluations.

A recent narrative review[15] underlines the challenges in interpreting the literature due to significant variability in cryoanalgesia techniques, including the probes used, the analgesic comparators, the type of surgery, and the primary outcomes studied. The majority of studies in thoracic surgery, mostly focussed on thoracotomy, report effective acute pain relief (−2.32 VNRS; 95% CI, −3.35 to −1.30), along with a reduction in opioid doses and decreased incidence of nausea (relative risk, 0.23; 95% CI, 0.06 to 0.95) compared with opioids alone.[6] When considering only thoracotomies, cryoanalgesia appears superior to intercostal blocks and opioid-based analgesia and potentially equivalent to epidural analgesia.[6]

The literature on cryoanalgesia in VATS lung resection is very limited. O'Connor and colleagues[9] conducted a retrospective comparison between patients who underwent cryoanalgesia and a historical cohort before its use. Cryoanalgesia performed at temperatures between −50 and −70 °C (AtriCure) on an average of 5 intercostal levels produced a reduction in opioid consumption and pain scores in both opioid-naïve and opioid-tolerant patients. However, their data included robotic-assisted surgeries and muscle-sparing thoracotomies, and analgesia was not standardized. Li and colleagues[10] published solely in Chinese and prospectively compared intercostal cryoanalgesia to intravenous opioids in 80 patients who underwent VATS. Pain scores at rest and with movement at 4 hours, day 1, day 2, and day 3 after surgery, morphine doses, and plasma concentrations of cortisol, C-reactive protein, and interleukin 6 were similar in the 2 groups. Nausea and vomiting were significantly higher in the intravenous analgesia group (36.1% vs 17.1%; $P < .05$). Weksler and colleagues[16] recently published a randomized controlled trial comparing intercostal cryoanalgesia in addition to the standard of care versus standard of care alone in 103 patients undergoing lung procedures (88.3% anatomic lung resections) using a robotic (76.7%) or VATS approach. The standard of care included internal intercostal nerve blocks from the second to the 10th intercostal spaces, as well as acetaminophen, gabapentin, tramadol, celecoxib, and ketorolac. Cryoanalgesia was performed at −80 °C using an AtriCure probe on 5 to 6 intercostal nerves at the levels corresponding to the thoracoscopic port sites, with a 2-minute freeze cycle per nerve. The total amount of morphine equivalents consumed during hospital stay (44.9 vs 38.4 mg; $P = .468$) and at 2 weeks (108.8 vs 95.2 mg; $P = .541$) was similar between groups. Pain scores also did not differ significantly. The authors did not report objective assessment of thoracic sensory loss. Neuropathic pain quantified with the leeds assessment of neuropathic symptoms and signs scale was higher in the cryoanalgesia group at 2 weeks (8 vs 13; $P = .019$), and the proportion of patients with a score ≥12 was significantly greater (32% vs 52%; $P = .045$). This result may reflect a higher incidence of neuropathic pain in the cryoanalgesia group. Numbness at the incision site is expected after successful cryoanalgesia but adds 3 points on the leeds assessment of neuropathic symptoms and signs scale, making interpretation of increased leeds assessment of neuropathic symptoms and signs scale score after cryoanalgesia ambiguous.

Cryoanalgesia demonstrated significant benefits for minimally invasive (bilateral VATS) Nuss procedures (minimally invasive pectus excavatum repair) used to correct pectus excavatum, including much lower opioid intake, shorter hospital stays, and lower pain scores compared with traditional pain management methods. Most studies utilized a transpleural approach with AtriCure probes on the intercostal nerves at temperatures between −60 and −70 °C for 2 minutes per nerve.[17-19] The mean age of these patients (15 years), differences in surgical indication, the extent of surgical trauma, and the nature of pain complicate comparison.

In the current study, the most plausible explanation for the lack of analgesic benefit from cryoanalgesia is a failure of the technique. Of the 39 patients who had cryoanalgesia and underwent Von Frey filament evaluation for hypoesthesia, only 2 exhibited hypoesthesia across all 7 targeted levels. This failure could occur if the nerve is not fully enveloped within the ice ball (cryoablation zone) or if the probe does not reliably achieve temperatures of −20 °C or lower, which are necessary for Wallerian degeneration. The current study used the Erbe probe, which is primarily designed for interventional bronchoscopy rather than cryoanalgesia. However, Erbe representatives report successful (unpublished) use of this probe for cryoanalgesia in minimally invasive pectus excavatum repair procedures. The probe used in the current study was selected because it had previously been used for subjectively successful intercostal analgesia following lung transplantation in our hospital. In most cases, intercostal nerves were visible transpleurally, and the probe was positioned precisely over the nerve. Compared with the 8- to 10-mm tip AtriCure probe[20] successfully used for minimally invasive pectus excavatum repair cryoanalgesia,[15,17-19] the Erbe probe's 2.4-mm tip may not effectively generate an ice ball large and cold enough to produce the desired reversible Wallerian degeneration. A larger tip is associated with the formation of a bigger ice ball,[21] increasing the likelihood of complete nerve coverage. Moreover, whereas visual confirmation of ice ball formation on the probe was observed, it does not ensure that the critical temperature of −20 °C was reached.[15] In vitro experiments using ballistic gelatin and

the Erbe probe revealed that the critical temperature of −20 °C was not reliably achieved at distances of 0.5 and 1.0 cm from the probe tip during the first 3 minutes of the freeze cycles.[22] Finally, 0.4 mL/kg (up to 40 mL) anesthetic solution was injected in the paravertebral space at the beginning of surgery. Although cryoanalgesia was performed approximately 3 cm lateral to the paravertebral space, partial intercostal spread of the anesthetic solution, particularly at the T5 level, was often observed. Local anesthetic may have increased the distance between the parietal pleura and intercostal nerves, potentially affecting the temperature achieved at the level of the intercostal nerves.

In ultrasound-guided transdermal cryoanalgesia, even with smaller tip sizes, the needle can be positioned much closer to the nerve than in transpleural techniques, allowing the ice ball to form directly next to the nerve. Precise positioning reduces the need for a larger ice ball and potentially improves the likelihood of successful nerve freezing.[21,23] Some thoracotomy cryoanalgesia studies described dissecting the parietal pleura to bring the probe closer to the intercostal nerve. This more invasive approach risks direct nerve injury or the double-crush phenomenon, potentially leading to chronic neuropathic pain after cryoanalgesia.[15] Currently, no studies have compared the in vivo reliability of different probes in consistently achieving the critical temperature for cryoanalgesia. Identifying the most effective probe for complete nerve encasement and reliable temperature achievement is crucial because an incomplete freeze will not result in analgesia and could even cause transitory hyperalgesia.[15] The current study found slightly, but significantly, more severe postoperative thoracic pain during cough at 1 month. At this time point, there was also a nonsignificant trend toward a higher incidence of persistent thoracic pain, greater thoracic pain at rest, increased ongoing opioid use, a higher rate of positive DN4 scores, and greater use of neuropathic pain medication in the cryoanalgesia group. This trend subsided by 3 months and 6 months. These findings may reflect the effects of an incomplete freeze or may simply be due to chance.

The current study has several strengths, including specifically addressing cryoanalgesia in VATS lung resection, rigorous and standardized cryoablation, multimodal analgesia and patient evaluation protocols, and no loss to follow-up. Objective assessment of the extent of thoracic sensory loss using the Von Frey filament made it possible to distinguish between ineffective cryoanalgesia and effective cryoanalgesia that does not reduce VATS pain. Moving forward, providing a detailed description of cryoanalgesia equipment and techniques, along with objective quantification of the resulting hypoesthesia and analgesia, will allow assessment of the effectiveness of specific cryoanalgesia approaches and simplify reconciling discrepancies in results.

CONCLUSIONS

The cryoanalgesia technique described in the present randomized clinical trial did not produce objective analgesia of the targeted hemithorax and showed no benefit compared with standard multimodal analgesia, including SPVB for managing acute or chronic thoracic pain following VATS lung resection. Significantly higher pain scores at cough 1 month postoperatively may have resulted from incomplete freeze and warrants caution. Future cryoanalgesia research should include detailed descriptions of equipment and technique and objective descriptions of sensory loss to more rapidly progress to the specific combination (if any) that effectively relieves both acute and chronic pain after VATS lung resection.

Conflict of Interest Statement

The authors reported no conflicts of interest.

The *Journal* policy requires editors and reviewers to disclose conflicts of interest and to decline handling or reviewing manuscripts for which they may have a conflict of interest. The editors and reviewers of this article have no conflicts of interest.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the corresponding author used ChatGPT in order to translate the study protocol from French to English and check for the spelling and structure of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Key Words: thoracic surgery, video-assisted thoracoscopic surgery, cryoanalgesia, pain

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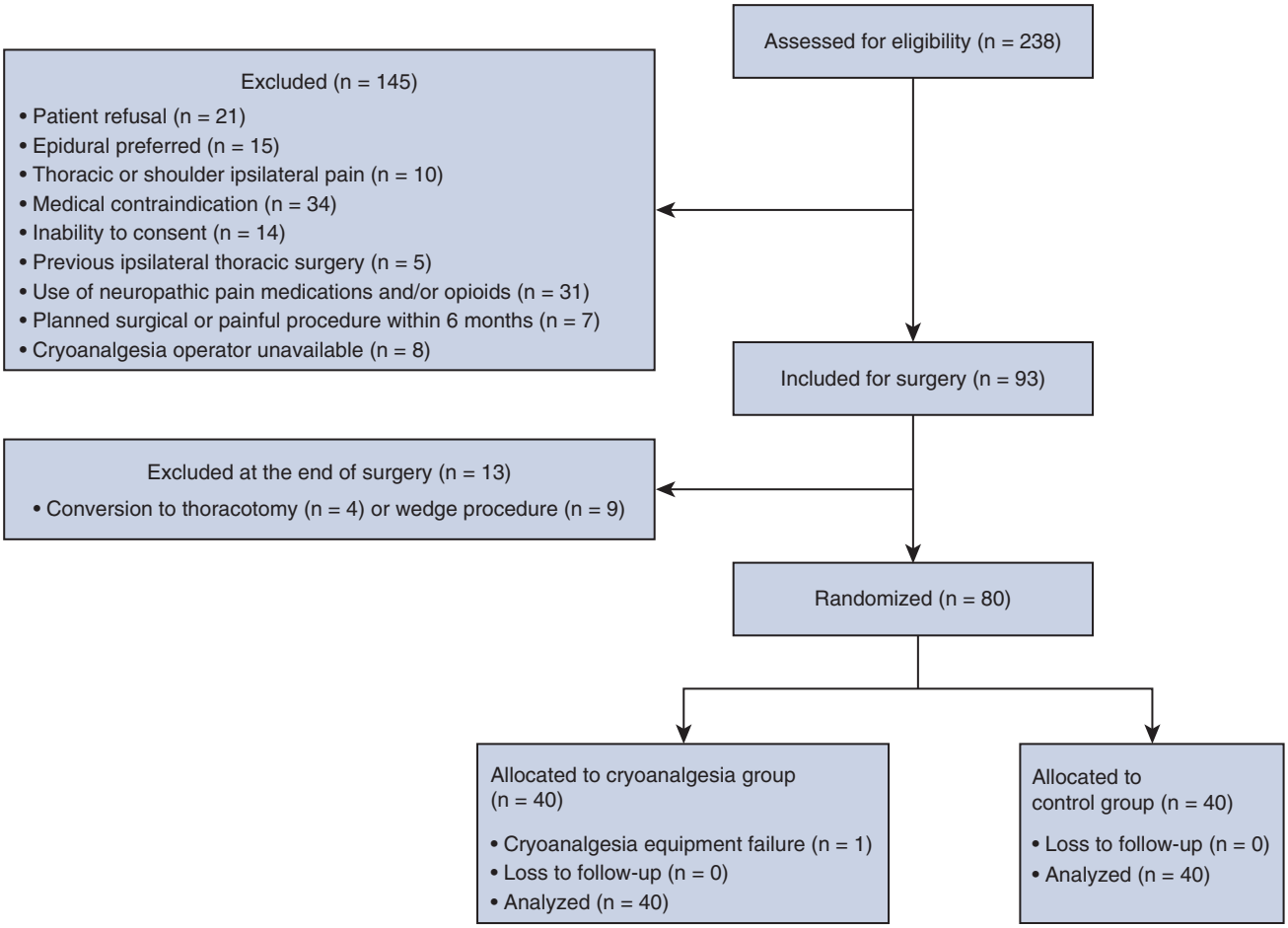

FIGURE E1. Consolidated Standards of Reporting Trails diagram.